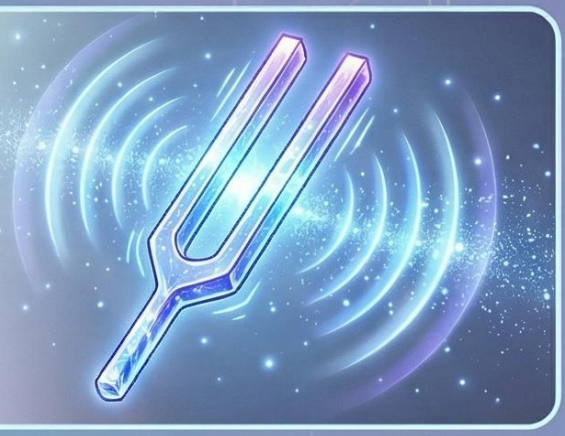
WHAT IS SOUND? (ध्वनी म्हणजे काय)

DEFINITION (व्याख्या)

Sound is a form of energy produced by vibrations.

ध्वनी म्हणजे कंपनांमुळे निर्माण होणारी ऊर्जा.

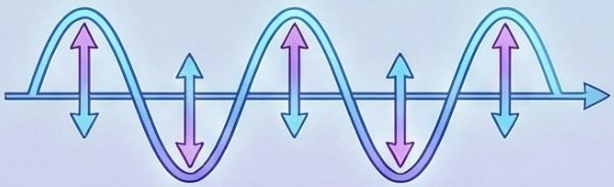


SOUND AS ENERGY + VIBRATIONS (ध्वनी = ऊर्जा + कंपन)

Example: Bell ringing, Guitar string
उदाहरण: घंटा, गिटारची तार

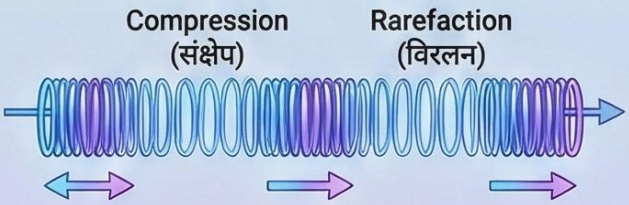
TYPES OF WAVES (लहरींचे प्रकार)

TRANSVERSE WAVES (आडव्या लहरी)

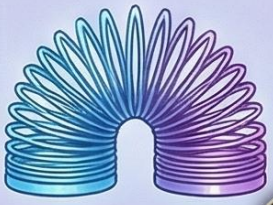


Particles vibrate perpendicular to wave direction.
कण लहरींच्या दिशेला लंब हालतात.

LONGITUDINAL WAVES (अनुदैर्घ्य लहरी)



Particles vibrate parallel to wave direction.
कण लहरींच्या दिशेनेच पुढे-मागे हालतात.



FREQUENCY & PITCH (वारंवारता आणि स्वरमान)

DEFINITION (व्याख्या)

Frequency is the number of vibrations (compressions and rarefactions) produced per second.
वारंवारता म्हणजे एका सेकंदात निर्माण होणाऱ्या संकोचन व विरलनांची संख्या.

SI Unit: **Hertz (Hz)**

$$\text{Frequency (f)} = \frac{\text{Number of vibrations}}{\text{Time}}$$

PITCH EXPLANATION (स्वरमान स्पष्टीकरण)

Pitch is the sensation of how high or low a sound appears to the human ear.
Pitch पूर्णपणे वारंवारतेवर अवलंबून असतो.

VISUAL COMPARISON (तुलना)

HIGH FREQUENCY → HIGH PITCH
Example: Bird chirping (पक्ष्यांचा किलबिलाट)

LOW FREQUENCY → LOW PITCH
Example: Lion's roar (सिंहाची डरकाळी)

TYPES OF SOUND (ध्वनीचे प्रकार)

INFRASONIC WAVES (अध्वनी)

Sound waves with frequency less than 20 Hz.
२० Hz पेक्षा कमी वारंवारतेचे ध्वनी.

Example: Elephant, Whale communication

AUDIBLE WAVES (श्राव्य ध्वनी)

Sound waves with frequency between 20 Hz and 20,000 Hz.
२० Hz ते २०,००० Hz मधील ध्वनी.

Example: Human speech, Music

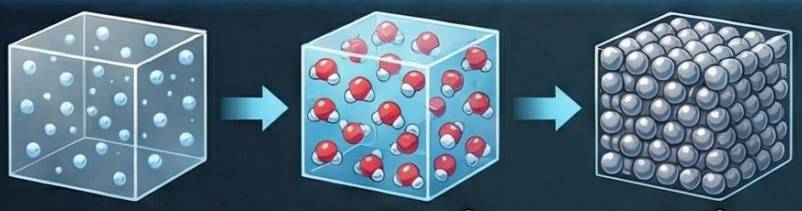
ULTRASONIC WAVES (अतिध्वनी)

Sound waves with frequency greater than 20,000 Hz.
२०,००० Hz पेक्षा जास्त वारंवारतेचे ध्वनी.

Example: Bat, Dog, SONAR

SPEED OF SOUND (ध्वनीचा वेग)

The speed of sound depends on the medium and temperature.
ध्वनीचा वेग माध्यम आणि तापमानावर अवलंबून असतो.



AIR (हवा) → WATER (पाणी) → STEEL (पोलाद)

Speed increases as particles are closer

When temperature increases, speed of sound in air increases.
तापमान वाढल्यास हवेतील ध्वनीचा वेग वाढतो.

IMPORTANT FORMULA:

Speed = Frequency × Wavelength
f Σ λ

DOPPLER EFFECT (डॉप्लर परिणाम)

Doppler Effect is the apparent change in frequency of sound due to relative motion between the source and the observer.

स्रोत आणि निरीक्षक यांच्या परस्पर हालचालीमुळे ध्वनीच्या वारंवारतेत होणारा भासमान बदल म्हणजे डॉप्लर परिणाम.

Proposed by **Christian Doppler** in 1842.

ख्रिश्चन डॉप्लर यांनी १८४२ मध्ये मांडले.

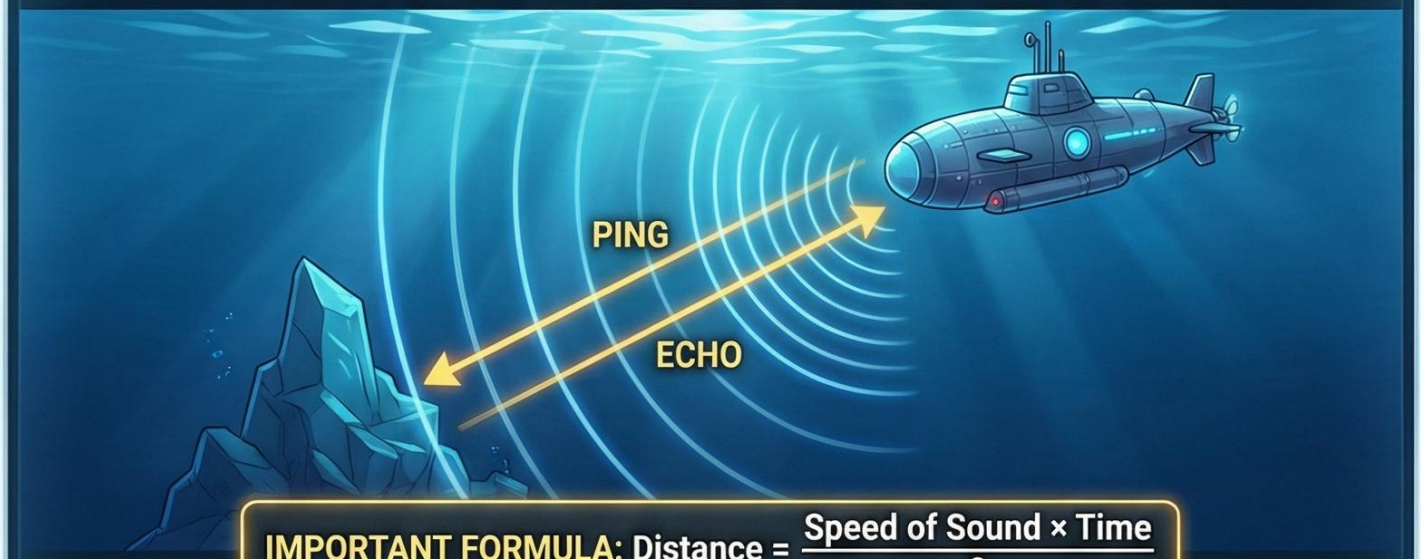
SOURCE APPROACHING OBSERVER → WAVES CLOSER → FREQUENCY INCREASES → HIGH PITCH
(जवळ येताना आवाज बारीक वाटतो)

SOURCE MOVING AWAY → WAVES SPREAD → FREQUENCY DECREASES → LOW PITCH
(दूर जाताना आवाज घोगरा वाटतो)

SONAR & ECHOLOCATION

SONAR (Sound Navigation and Ranging) is a technique that uses sound waves to detect underwater objects and measure distance.

SONAR म्हणजे पाण्याखालील वस्तू शोधण्यासाठी व अंतर मोजण्यासाठी ध्वनीलहरींचा वापर करणारी पद्धत.



IMPORTANT FORMULA: $\text{Distance} = \dfrac{\text{Speed of Sound} \times \text{Time}}{2}$

Echolocation note: Dolphins and bats use natural SONAR called Echolocation.

डॉल्फिन आणि वटवाघळे Echolocation वापरतात.

QUICK EXAM TIPS & MEMORY HOOKS (परीक्षेसाठी महत्त्वाचे मुद्दे)

High Contrast, Exam Points, Last-Minute Revision

CARD 1

SOUND BASICS

- ✓ Sound is a form of **energy** produced by **vibrations**.
- ✓ ध्वनी म्हणजे **कंपनामुळे** निर्माण होणारी **ऊर्जा**.
- ✓ Example: Bell, Guitar string

CARD 2

MEDIUM REQUIREMENT

- ✓ Sound cannot travel in **vacuum**.
- ✓ ध्वनीला **माध्यम** आवश्यक असते.
- ✓ Exception reminder: Light waves do not need medium.

CARD 3

FREQUENCY & PITCH

- ✓ **Frequency** decides **Pitch**.
- ✓ **Pitch वारंवारतेवर** अवलंबून असतो.
- ✓ **High frequency** → High pitch
- ✓ **Low frequency** → Low pitch

CARD 4

INTENSITY

- ✓ **Intensity** is the **loudness** of sound.
- ✓ **तीव्रता** म्हणजे आवाज किती **मोठा वाटतो**.
- ✓ Measured in: **Decibel (dB)**

CARD 5

SPEED OF SOUND

- ✓ **Speed** depends on **medium** and **temperature**.
- ✓ **वेग माध्यम** व **तापमानावर** अवलंबून.
- ✓ Order reminder: **Air < Water < Steel**

CARD 6

$$v = f \times \lambda$$

IMPORTANT FORMULA

- ✓ **Speed = Frequency × Wavelength** ($v = f \times \lambda$)

CARD 7

DOPPLER EFFECT

- ✓ **Relative motion** causes change in **frequency**.
- ✓ **स्रोत** व **निरीक्षकाच्या हालचालीमुळे वारंवारता** बदलते.

CARD 8

SONAR

- ✓ Used to measure **distance underwater**.
- ✓ **पाण्याखाली अंतर मोजण्यासाठी** वापर.